

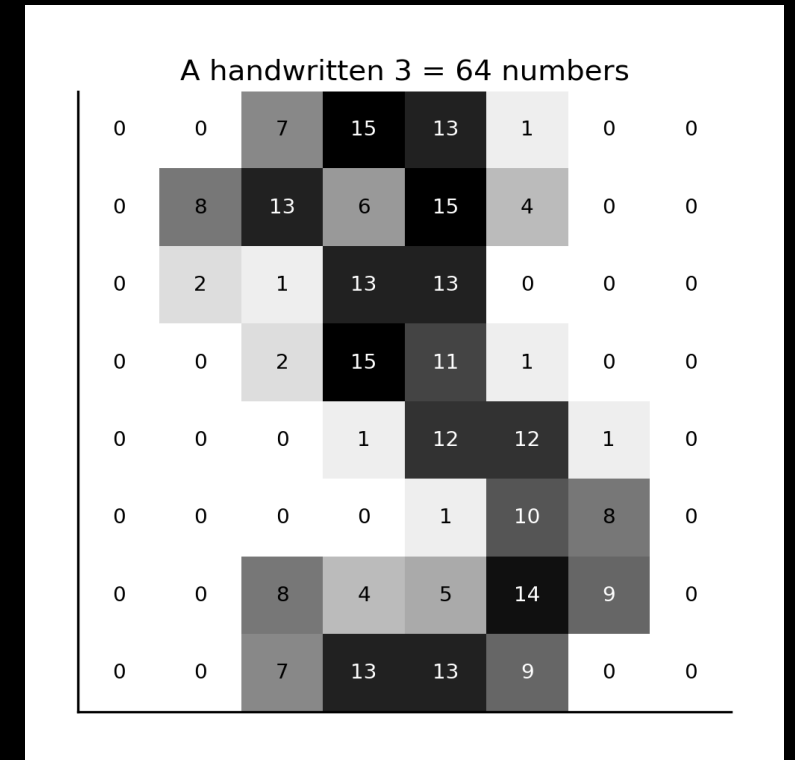
Introduction to Machine Learning

What ML is, how machines learn from data, and where it is used

Behnam Kiani Kalejahi

-

Fall Semester 2026–2027 · Module 1: Foundations



How can a computer learn to recognise this “3”?

Welcome to Machine Learning



Your lecturer

- Behnam Kiani Kalejahi
- Kiani-bme@hotmail.com
- Office hours: by appointment



How we learn

- Offline, in-person classes
- Traditional lecture + live demo
- Computer laboratory every week
- Python · Jupyter · Google Colab



What you need

- Computer Programming (Python basics)
- Basic Probability and Statistics
- A Google account (for Colab)
- A GitHub account (for projects)
- Curiosity and regular practice

The course at a glance

10

credits

30 h

lectures

60 h

lab classes

15

weeks + exam week

4

independent-work tasks (IWS)

7

consultations (IWST)

Course learning outcomes: by the end you will be able to...

L01



Build

Use Python notebooks to prepare data, train simple ML models and produce basic predictions.

L02



Choose & evaluate

Pick a suitable model for regression, classification or clustering and evaluate it with the right metrics.

L03



Communicate

Explain the workflow, results and limitations clearly in a short report and presentation.

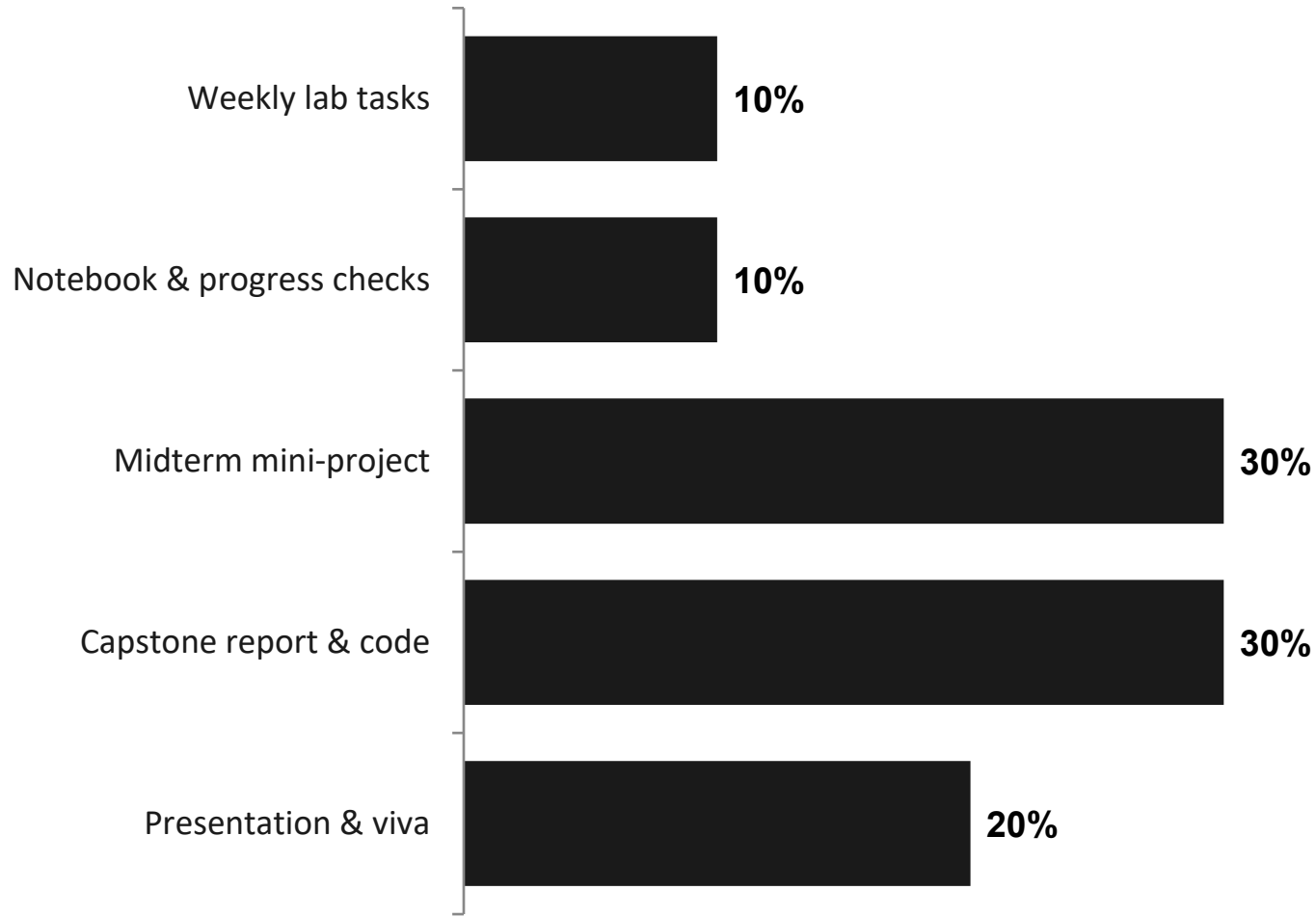
L04



Collaborate

Work in a pair or small group and complete a simple ML project on time.

How you will be assessed



80% = projects

Midterm mini-project, capstone report & code, and presentation & viva make up most of your grade.



No written exam

The final control is your capstone viva in the final exam week. Deadlines are strict; late work loses points.

Your 15-week roadmap

MODULE 1

Foundations

W1 Introduction to ML (today)

W2 Python & pandas basics

W3 Preparing data

W4 Linear regression · MAE/MSE/RMSE

W5 Logistic regression · accuracy, F1

MODULE 2

Core models & clustering

W6 k-Nearest Neighbours

W7 Midterm mini-project demo

W8 Decision trees

W9 Naive Bayes

W10 Model comparison

W11 k-Means clustering

MODULE 3

Pipelines & capstone

W12 Simple ML pipelines

W13 Final project planning

W14 Lab support & rehearsal

W15 Capstone submission

Exam week: capstone viva

Today's learning objectives

1

Define machine learning and contrast it with traditional programming

2

Use the core vocabulary: features, labels, model, training, prediction

3

Distinguish supervised, unsupervised and reinforcement learning

4

Recognise real applications and judge when ML is (not) appropriate

5

Outline the seven steps of a typical ML workflow

6

Name the Python tools we will use all semester

Agenda

01



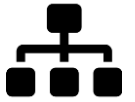
What is machine learning?

02



How machines learn

03



Types of machine learning

04



ML in the real world

05



Workflow, limits & ethics

06



Tools & what's next

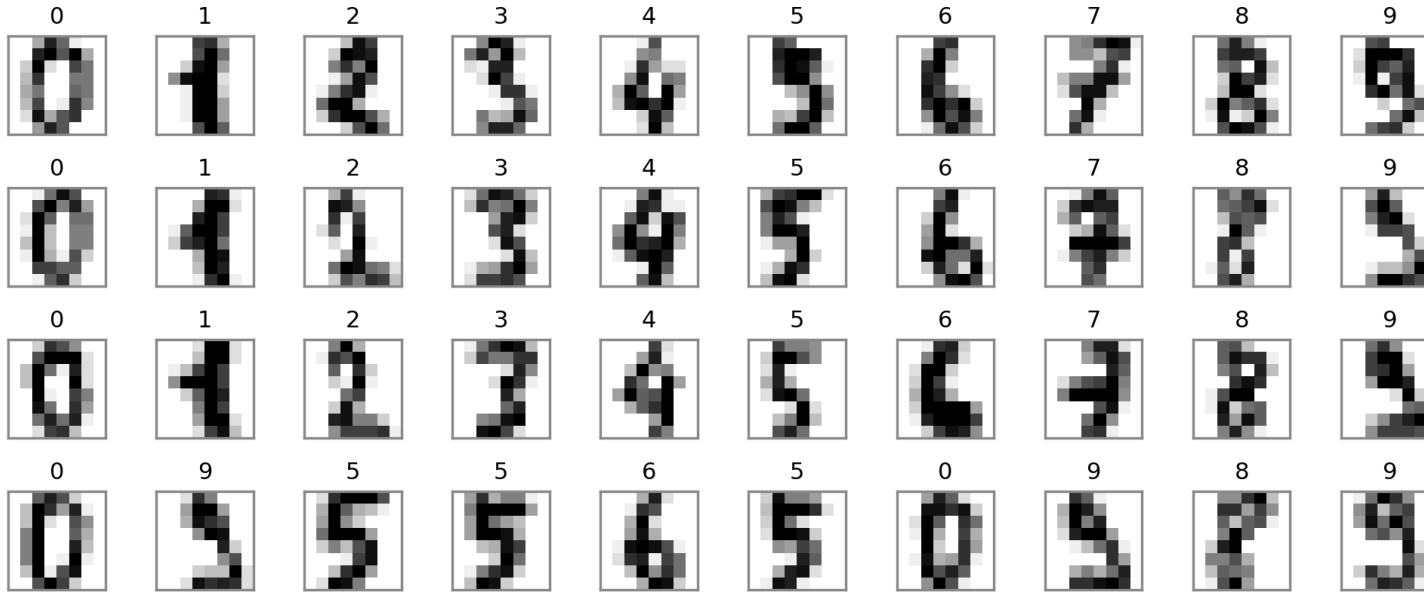
01

What is machine learning?

From hand-written rules to rules learned from data



Warm-up: write a program that recognises a “3”



Real 8×8 handwritten digits from the scikit-learn digits dataset (1,797 images).

Examples are easy to collect. Rules are hard to write.

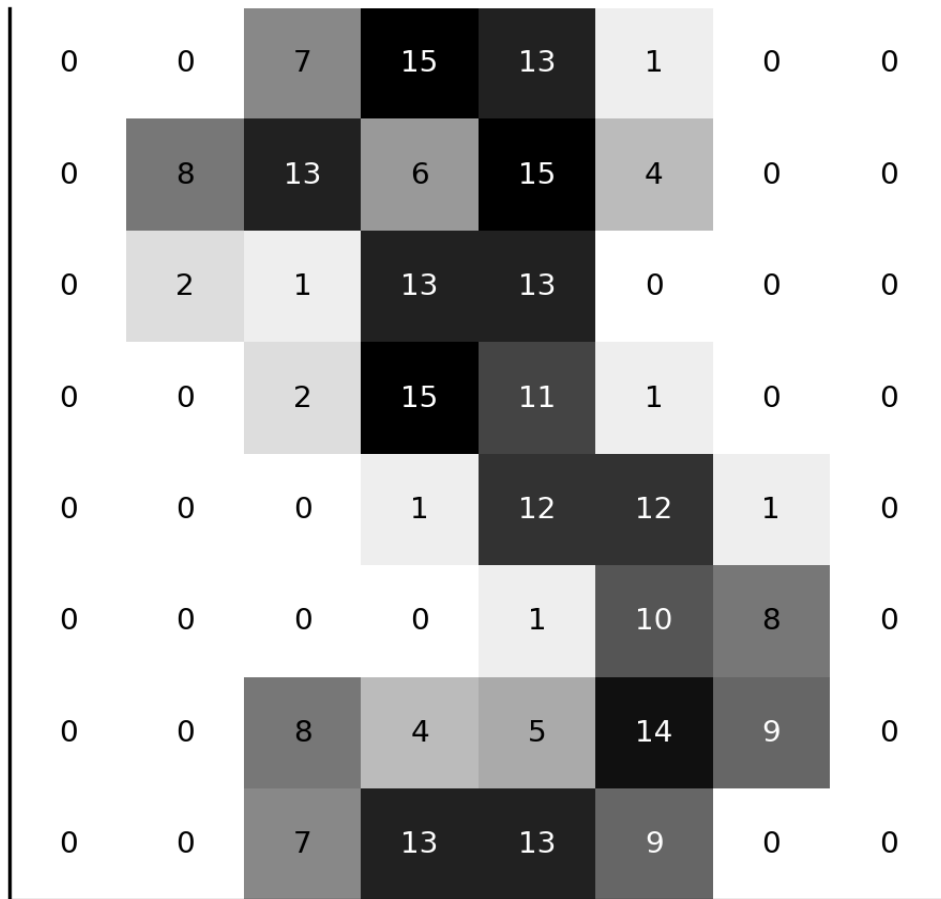


Discuss (1 minute)

- What rules would you write in `is_three(image)`?
- Does your rule work for slanted or thick 3s?
- How would you tell a 3 from an 8 or a 5?
- How many rules would you need for all ten digits?

Why hand-written rules fail

A handwritten 3 = 64 numbers



Too many inputs

64 pixels $\times$ 17 grey levels already gives an astronomically large number of possible images.



Too much variation

Slant, thickness, size and style differ from writer to writer — and from day to day.

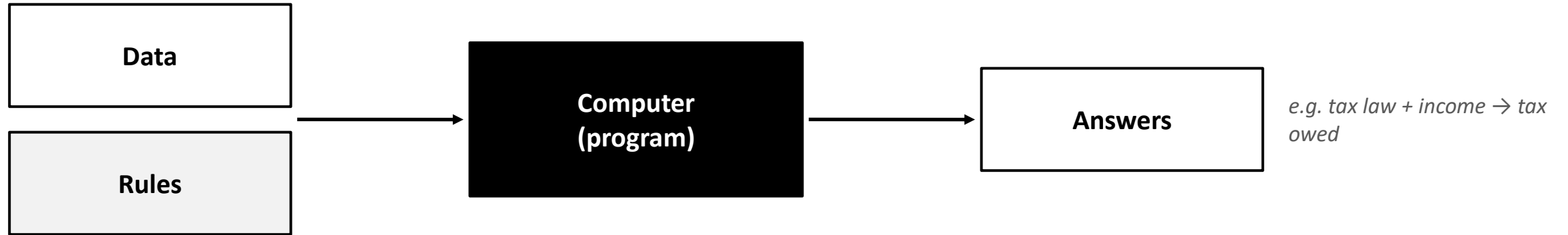


The ML answer

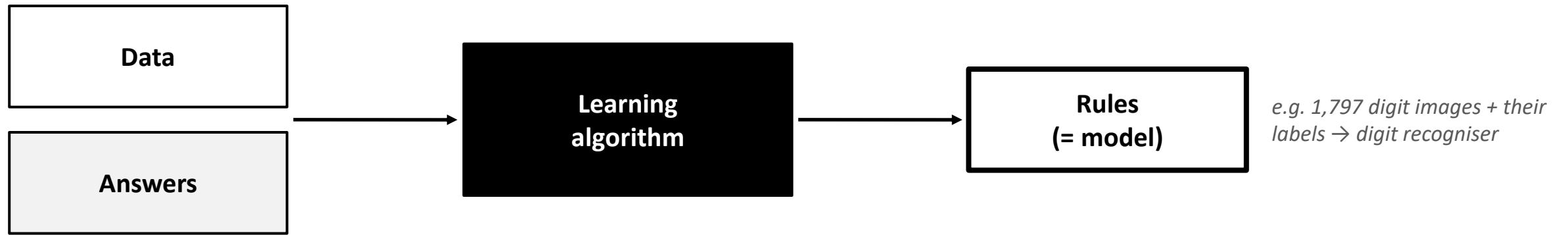
Show the computer thousands of labelled examples and let it discover the rule itself.

Traditional programming vs machine learning

TRADITIONAL PROGRAMMING



MACHINE LEARNING



A classic definition (Arthur Samuel, 1959)



Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.

— Arthur Samuel's idea, as it is commonly summarised (1959)



The checkers player

Samuel's program played thousands of games against itself and learned which positions lead to wins.



Key phrase

“Not explicitly programmed”: we write code that learns, not code that contains the answer.

A precise definition: Task, Experience, Performance

T

Task

What should the system do?

Spam example: *Classify an e-mail as spam or not spam*

E

Experience

What data does it learn from?

Spam example: *10,000 e-mails already labelled by users*

P

Performance

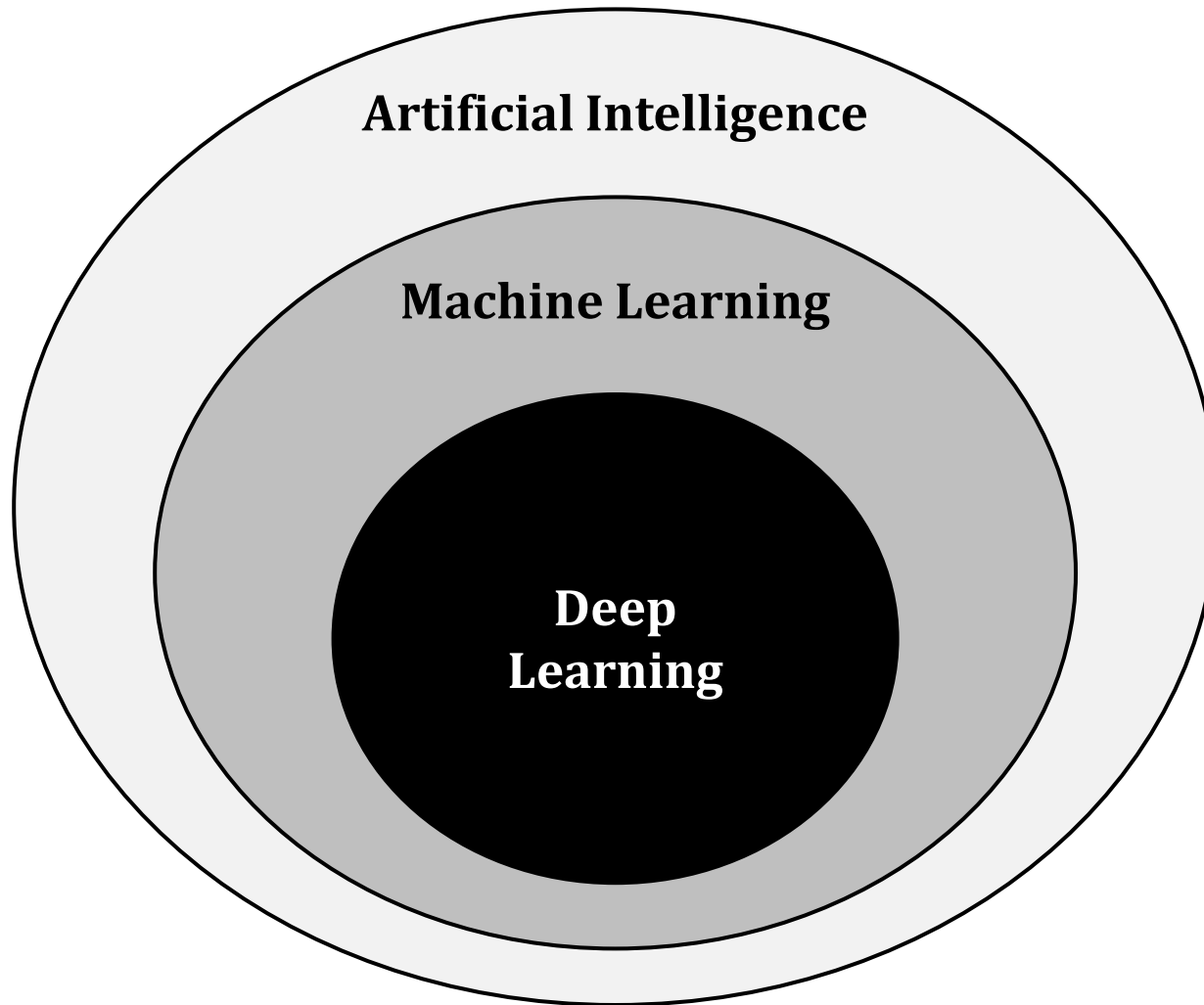
How do we measure success?

Spam example: *% of e-mails classified correctly on new data*

Applying T, E and P to real problems

Problem	Task (T)	Experience (E)	Performance (P)
Playing Go	Choose the next move	Games played, including against itself	% of games won
House prices	Predict a sale price	Records of past house sales	Average error in \$ (MAE)
Chest X-ray screening	Flag images showing disease	X-rays labelled by radiologists	Recall: % of true cases caught
Customer groups	Find groups of similar customers	Purchase histories (no labels)	Cluster quality / business usefulness
Handwritten digits	Recognise digits 0–9	1,797 labelled digit images	Accuracy on unseen images

AI, machine learning and deep learning



Artificial Intelligence

Any technique for intelligent behaviour — including rule-based expert systems, search and planning.

Machine Learning

Learns behaviour from data. Regression, k-NN, trees, Naive Bayes, k-means — this course.

Deep Learning

Many-layered neural networks: image & speech recognition, large language models.

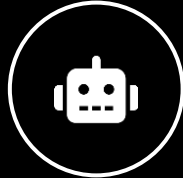
ML, statistics and data science: close relatives



Statistics

Goal: understand & infer

- Is this effect real?
- How certain are we?
- Simple, interpretable models
- Hypothesis tests, confidence intervals



Machine Learning

Goal: predict accurately

- How well will it work on new data?
- Often more complex models
- Train/test evaluation
- Automation at scale



Data Science

Goal: data → decisions

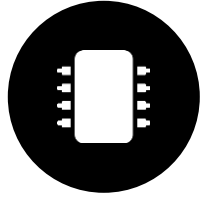
- Uses statistics and ML
- Data cleaning & engineering
- Visualisation
- Communicating results

Why is machine learning booming now?



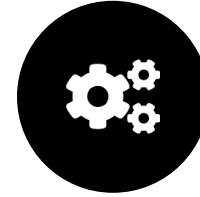
Data

Web, phones, sensors and digital records create huge labelled and unlabelled datasets.



Compute

GPUs and cloud platforms make heavy calculation cheap — Colab gives you one free.



Algorithms

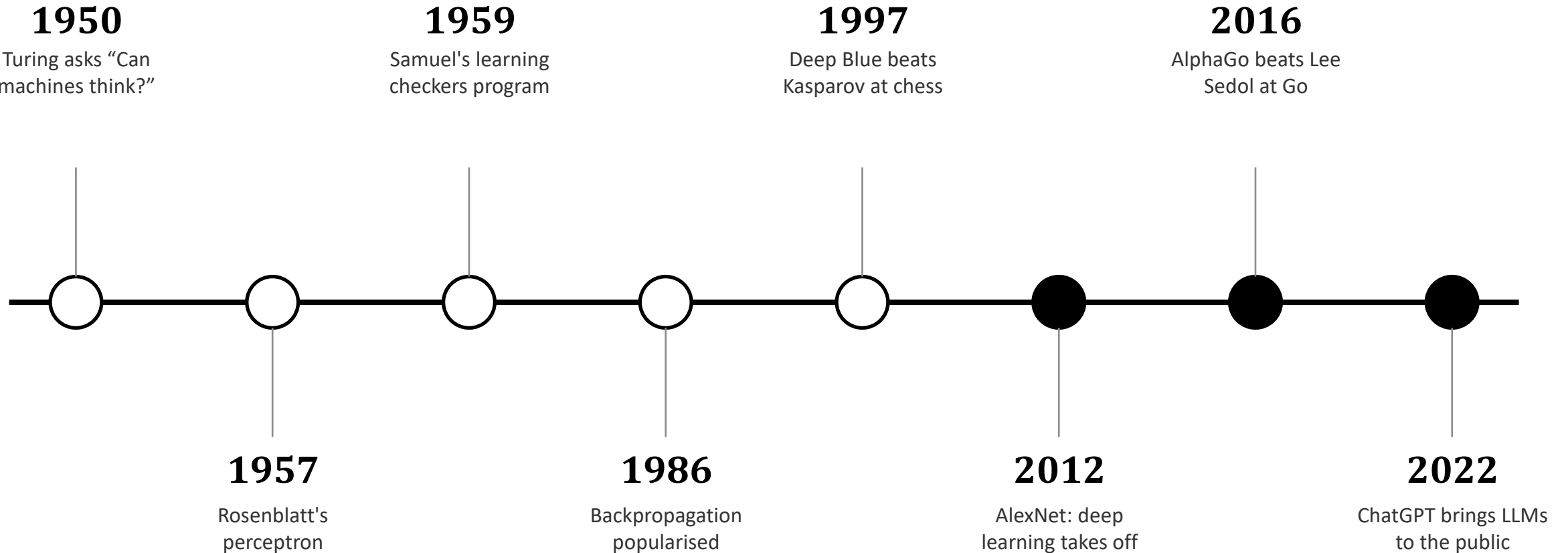
Better methods for training models, especially deep neural networks.



Open source

scikit-learn, PyTorch, TensorFlow, Kaggle: powerful tools for everyone.

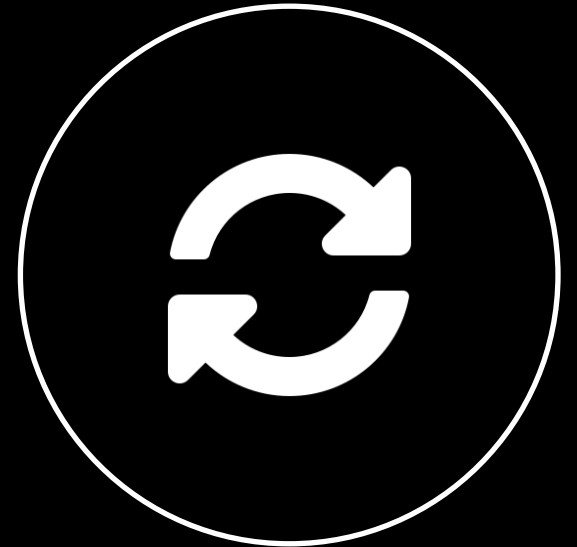
A brief history of machine learning



02

How machines learn

Data, features, labels, models — and the training loop



Anatomy of a dataset

← features X →

label y

Size (m ²)	Rooms	Age (yrs)	District	Price (\$k)
72	2	15	Chilonzor	68
110	3	4	Yunusobod	124
55	1	30	Sergeli	41
140	4	8	Mirzo Ulugbek	162
95	3	20	Chilonzor	?

Illustrative data, invented for teaching.



One row = one sample

Here: one house.



Columns = features

Numerical (size) or categorical (district).



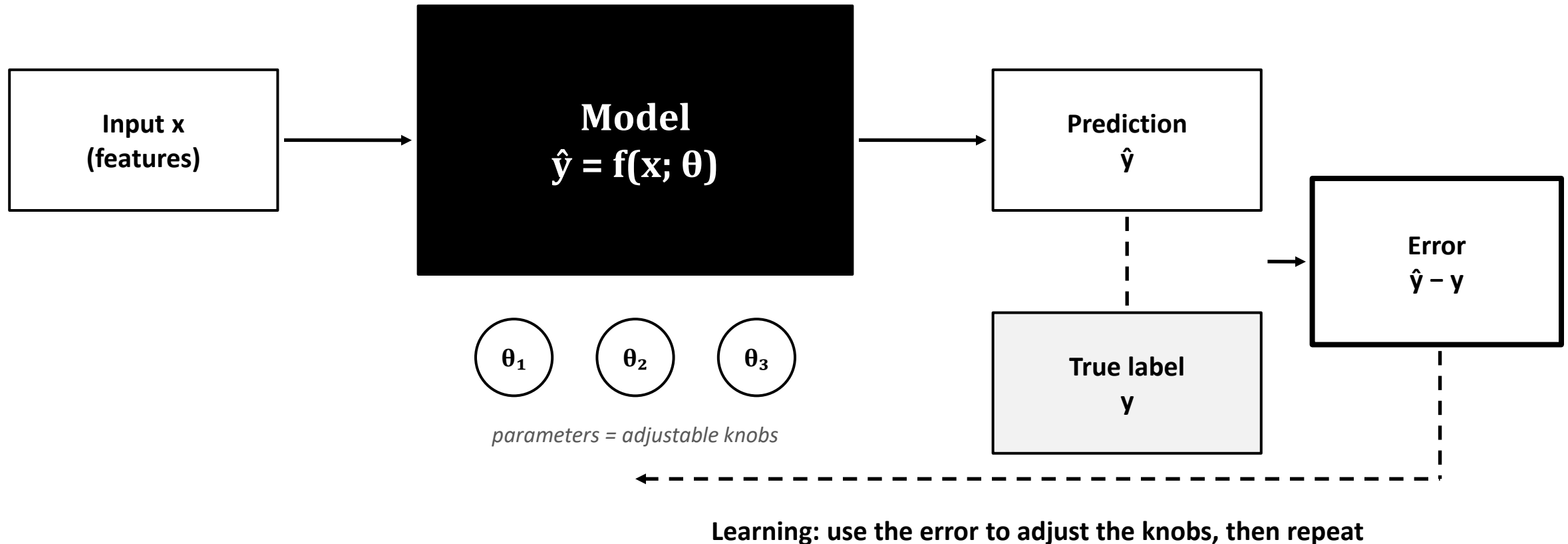
Target = label y

The value to predict. Last row: unknown!

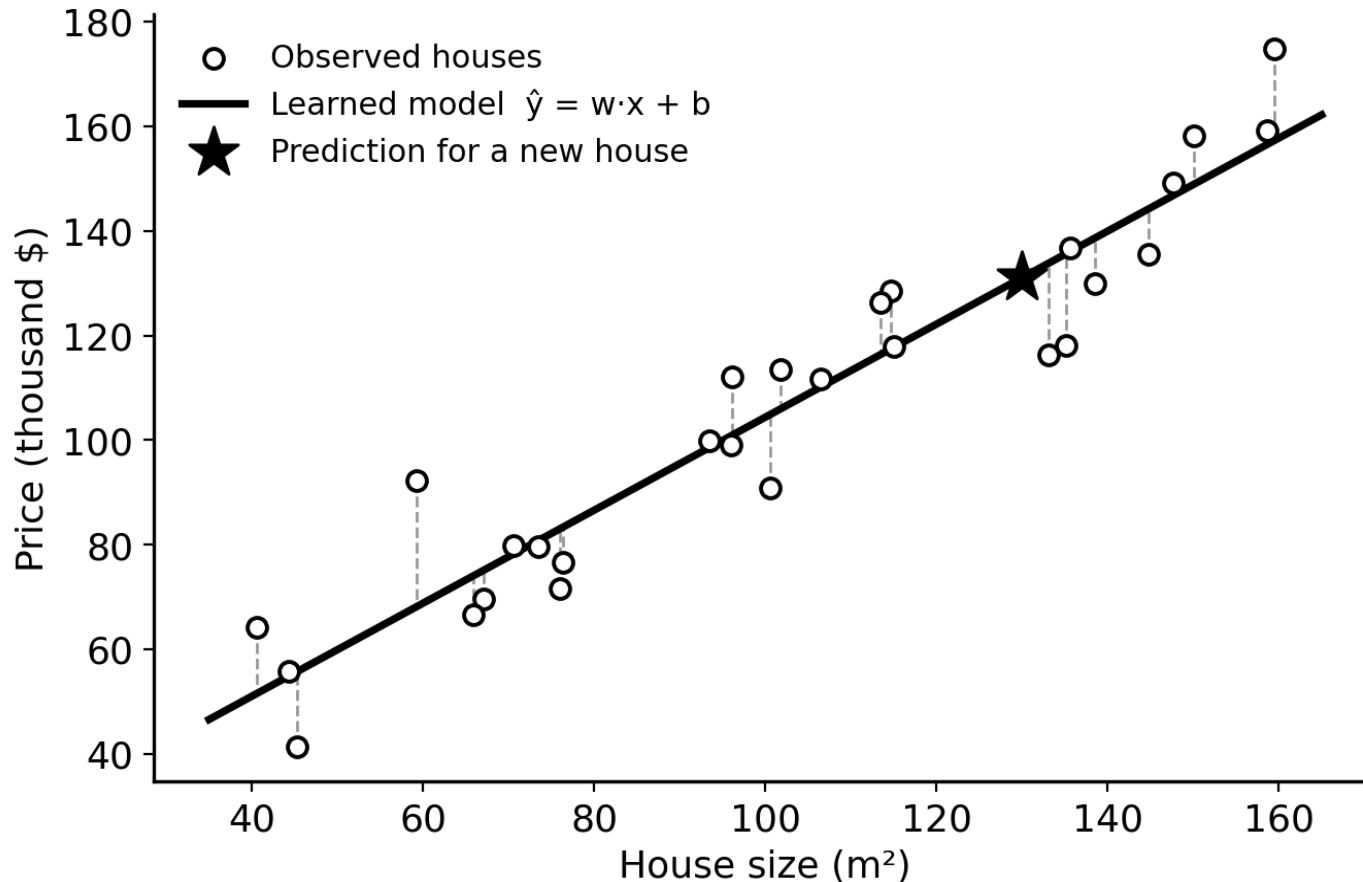
Key vocabulary you will use every week

Term	Meaning	House-price example
Sample / instance	One example (one row)	One house
Feature (X)	An input variable (a column)	Size, rooms, age, district
Label / target (y)	The output we want to predict	Price
Model	The learned rule f that maps X to y	$\text{price} \approx 0.9 \times \text{size} + 15$
Parameters	Numbers learned from data	0.9 (slope) and 15 (intercept)
Hyperparameters	Settings we choose before training	k in k -NN, max depth of a tree
Training	Fitting parameters to data	<code>model.fit(X_train, y_train)</code>
Prediction	Using the model on new inputs	<code>model.predict(X_new)</code>
Loss / error	How wrong the predictions are	Average error of \$12k

A model is a function with adjustable knobs



Learning = reducing error



Synthetic house data. Dashed lines = residuals (errors).

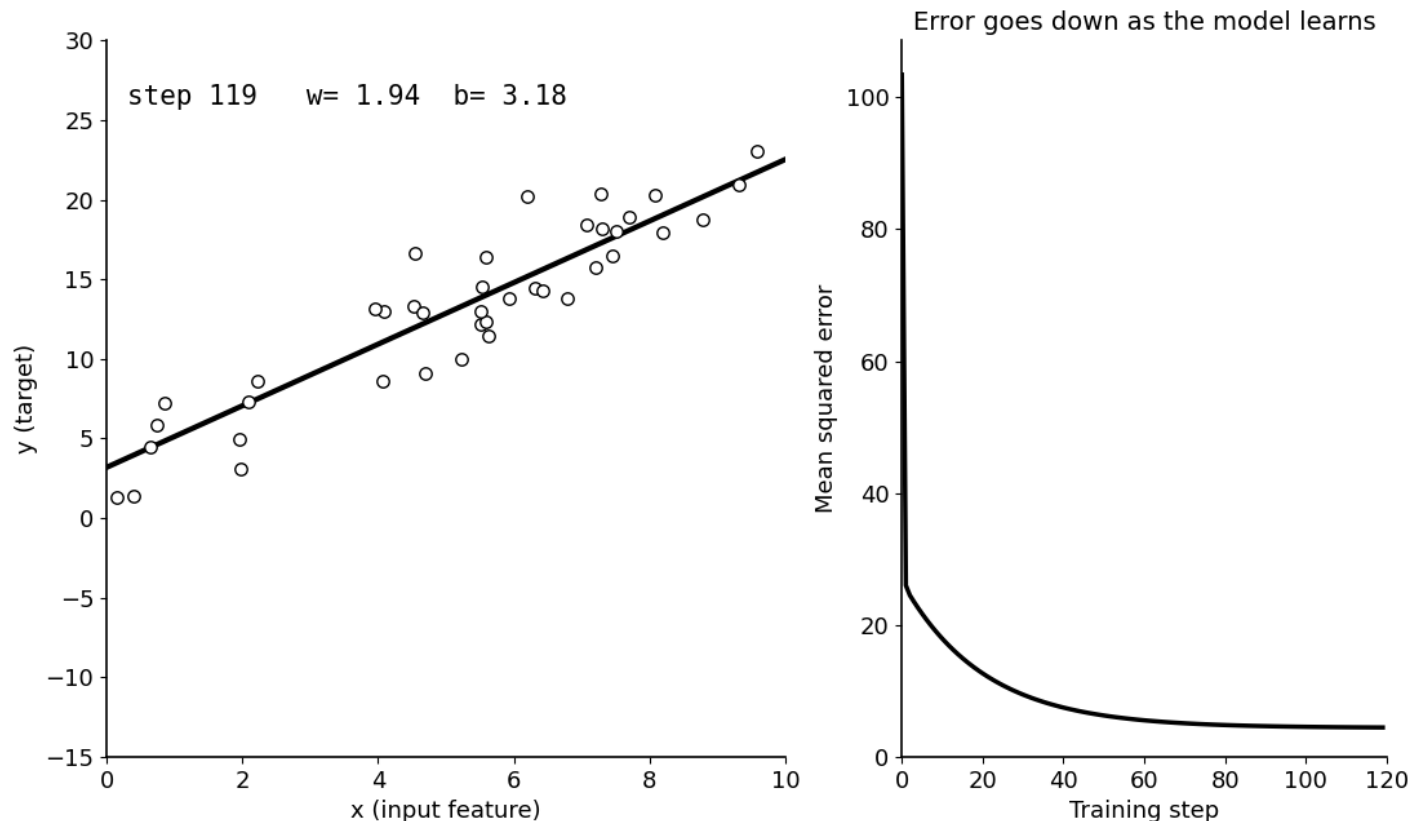
Mean Squared Error

$$\text{MSE} = (1/n) \cdot \sum (y_i - \hat{y}_i)^2$$

- **Square** so + and – errors don't cancel, and big errors count more.
- **Minimise** training searches for the w and b with the lowest MSE.
- **Predict** the trained line estimates the price of a new house (★).

Video: watch a model learn

Watch a model learn: fitting a line by reducing error

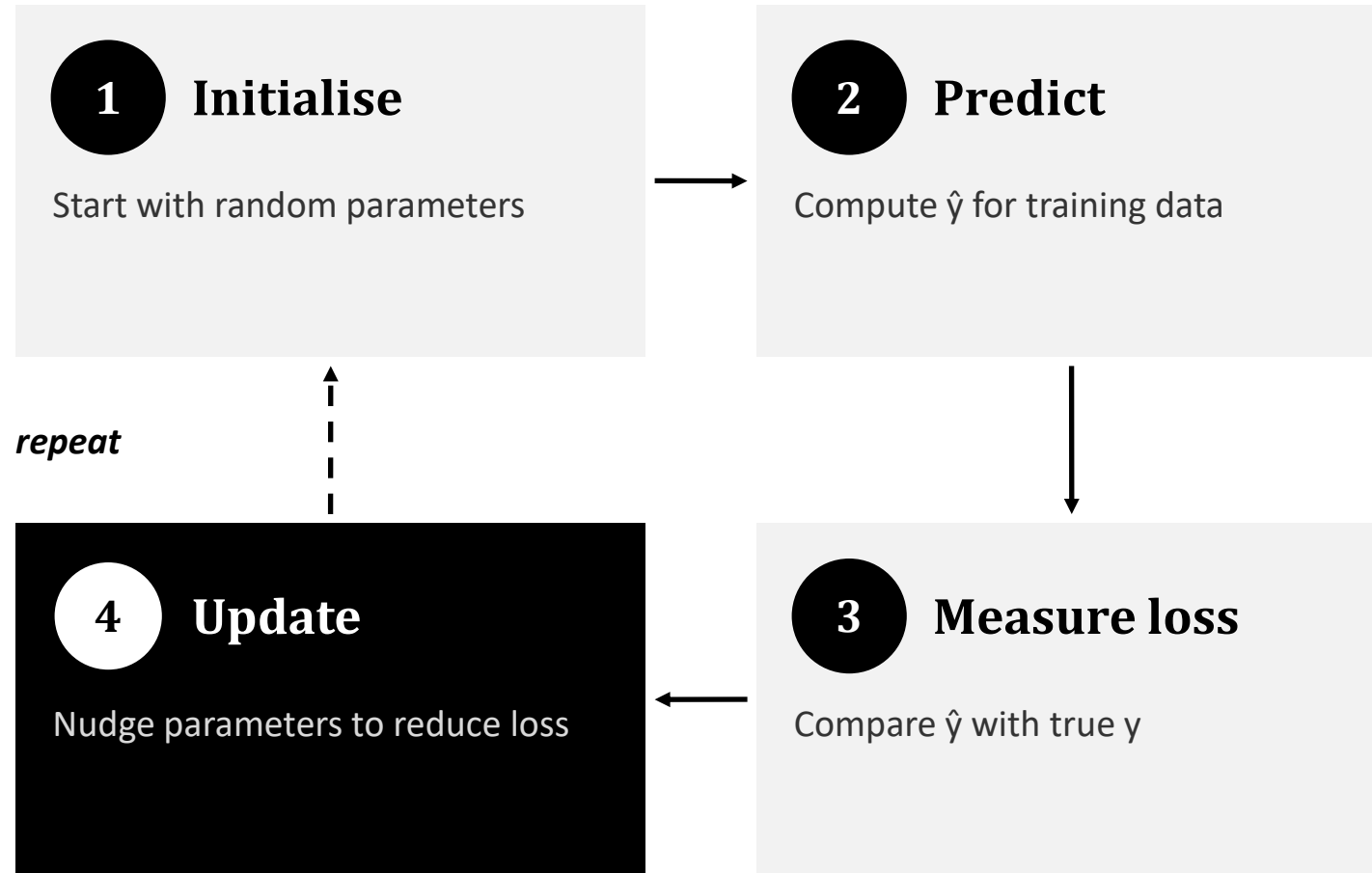


What to watch for

- Start: random parameters, a poor line
- Each step moves w and b a little
- The error curve falls fast, then flattens
- The final line fits the trend, not every point

Click to play (≈ 7 s). Animation generated for this lecture with Python/matplotlib.

The training loop



In scikit-learn the whole loop is one line:

```
model.fit(X, y)
```

Understand what happens inside it — that is what this course is about.

Training data vs test data

Full dataset, shuffled randomly, then split once

Training set $\approx 80\%$ — the model learns from this

**Test $\approx 20\%$
(locked away)**



Exam analogy

Practising with the real exam questions proves memory, not understanding. A fair exam uses unseen questions — so does a fair model evaluation.



Data leakage

Any information from the test set that sneaks into training makes the score look better than reality. Covered in the Week 3 lab.

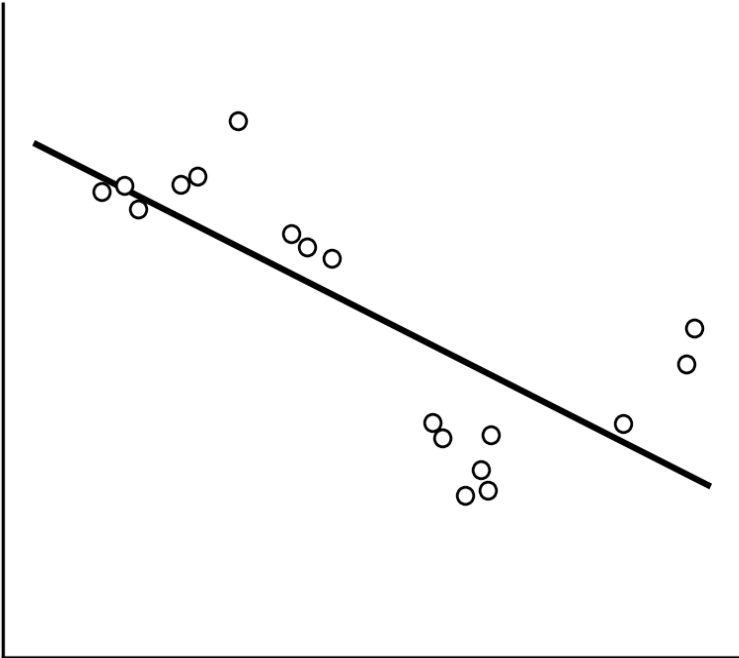
In code

```
from sklearn.model_selection
import train_test_split

X_train, X_test, y_train, y_test =
    train_test_split(X, y,
                    test_size=0.2,
                    random_state=42)
```

Generalisation: underfitting vs overfitting

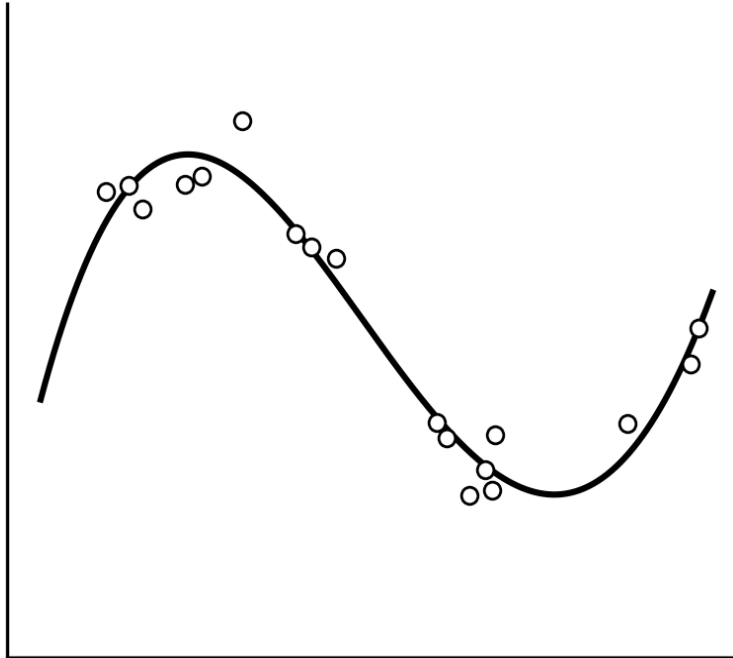
Underfitting (too simple)



Underfitting

Didn't study: misses even the obvious pattern.

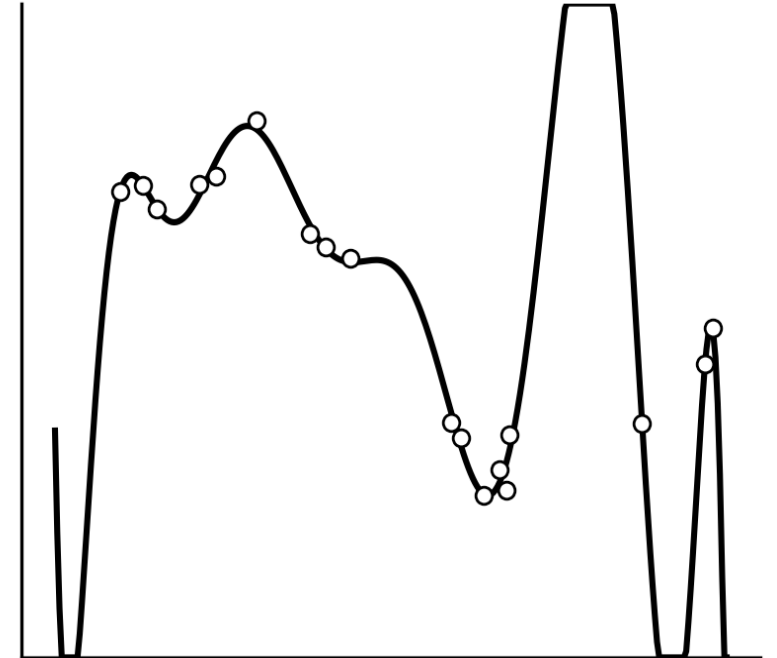
Good fit (generalises)



Good fit

Understood the ideas: works on new questions.

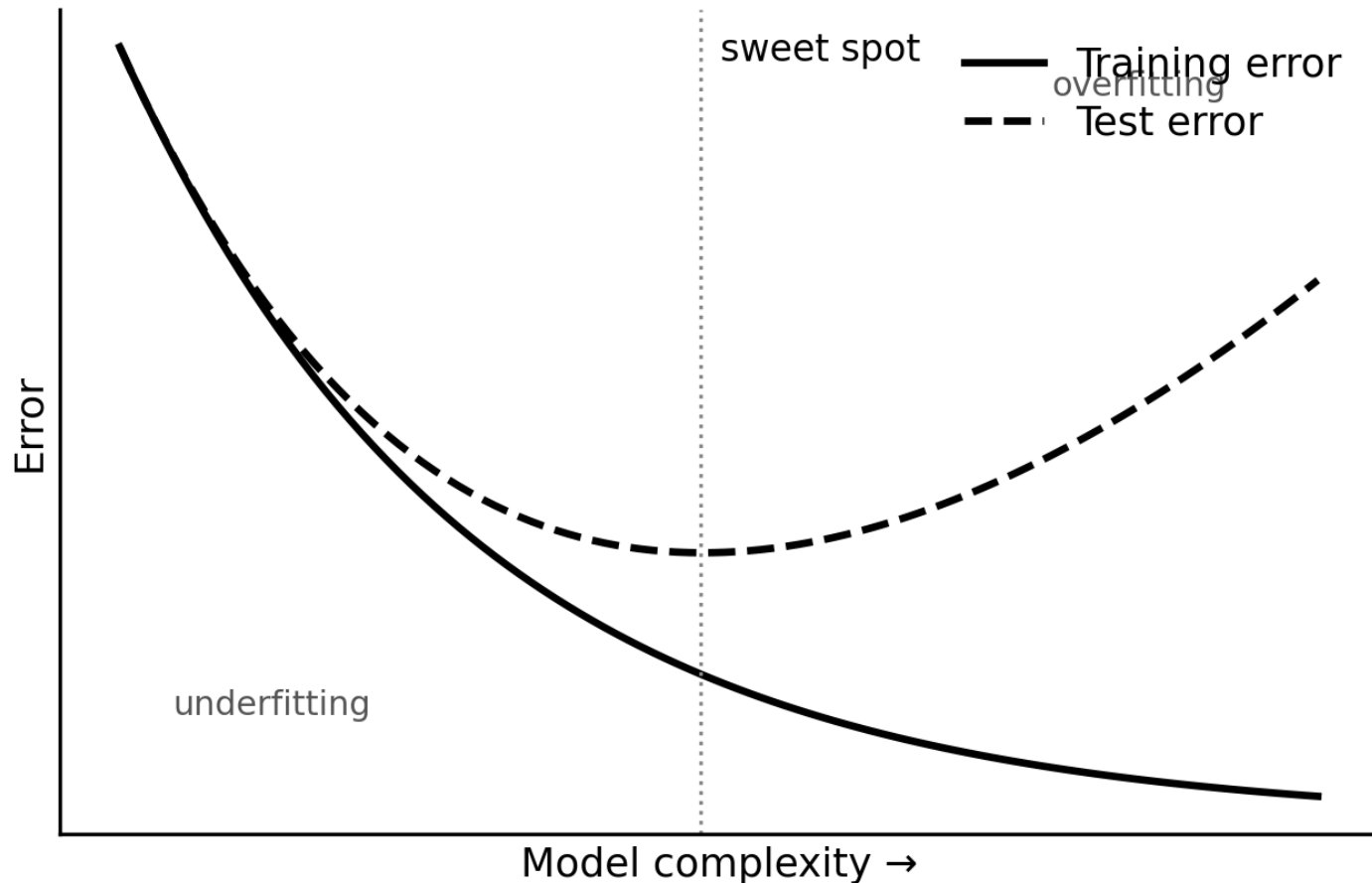
Overfitting (memorises)



Overfitting

Memorised last year's answers: fails on new questions.

The complexity trade-off



Illustrative curves showing the typical shape of training and test error.

Left side

Both errors high → underfitting. Use a more flexible model or better features.

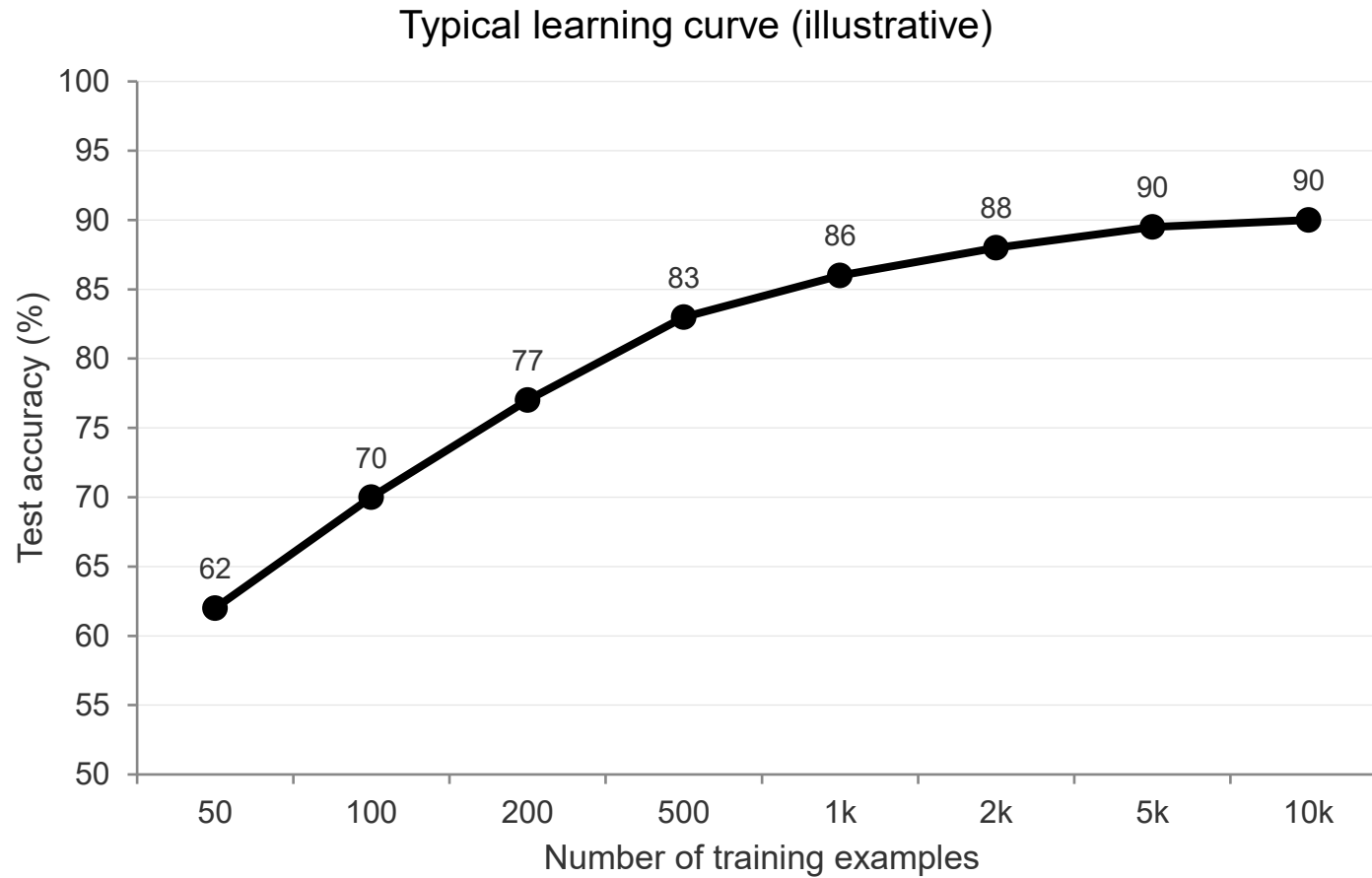
Sweet spot

Lowest test error → the model generalises best.

Right side

Training error tiny, test error rising → overfitting. Simplify or get more data.

More (good) data usually helps



Fast gains, then a plateau

Early examples teach a lot; later ones add less.



Quality beats quantity

Wrong labels teach wrong rules.



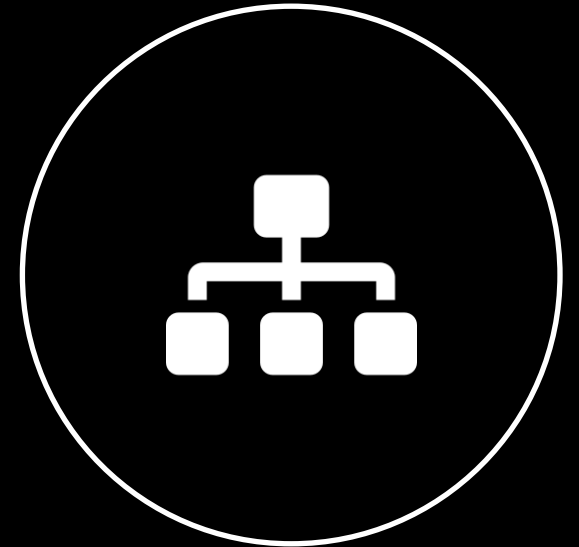
Representative data

Train on data that looks like the real world you'll predict on.

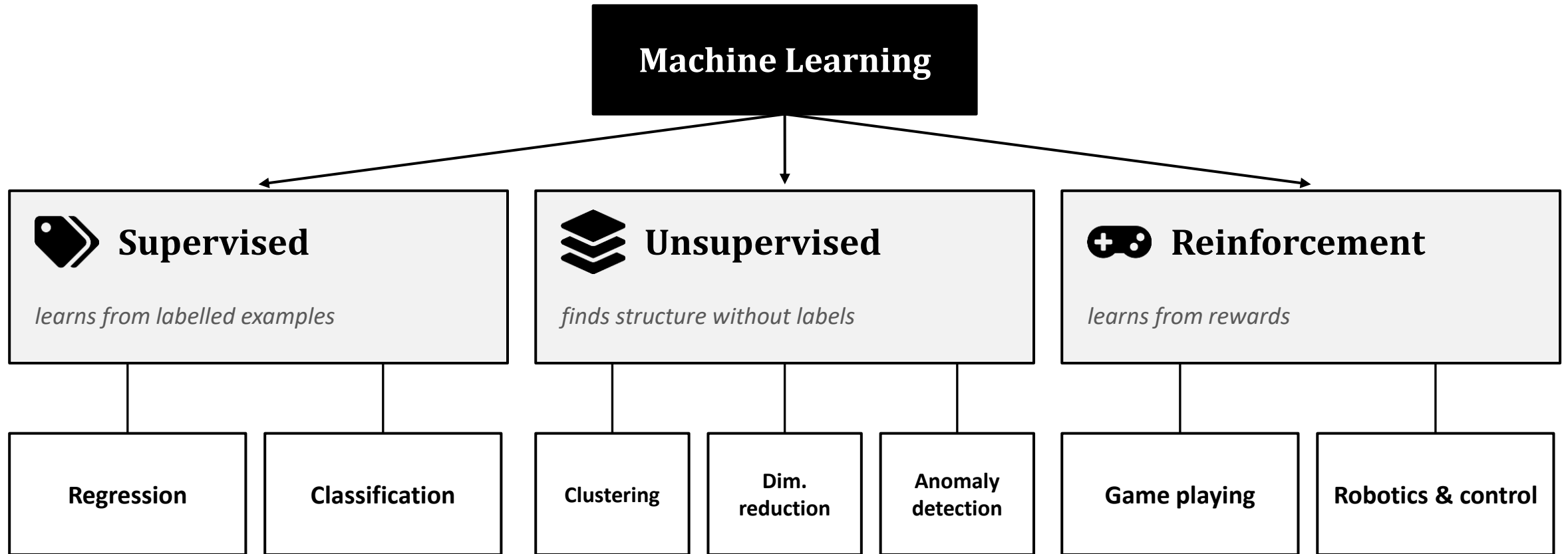
03

Types of machine learning

Supervised, unsupervised and reinforcement learning

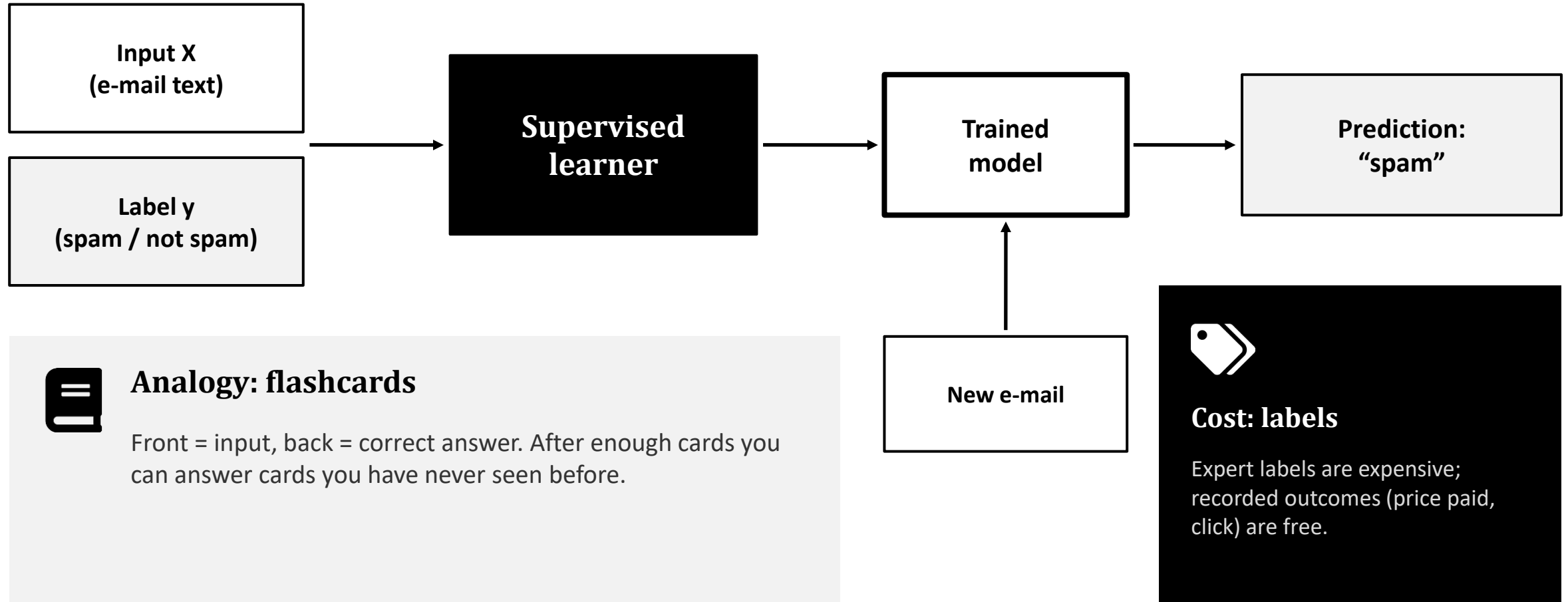


The three main types of machine learning



This course: supervised learning (Weeks 4–10) and k-means clustering (Week 11).

Supervised learning: learning with a teacher



Regression vs classification



Regression

Predict a **NUMBER**

- House price in \$
- Tomorrow's temperature
- Monthly energy use (kWh)
- A student's exam score

Model: linear regression

Metrics: MAE · MSE · RMSE (Week 4)



Classification

Predict a **CATEGORY**

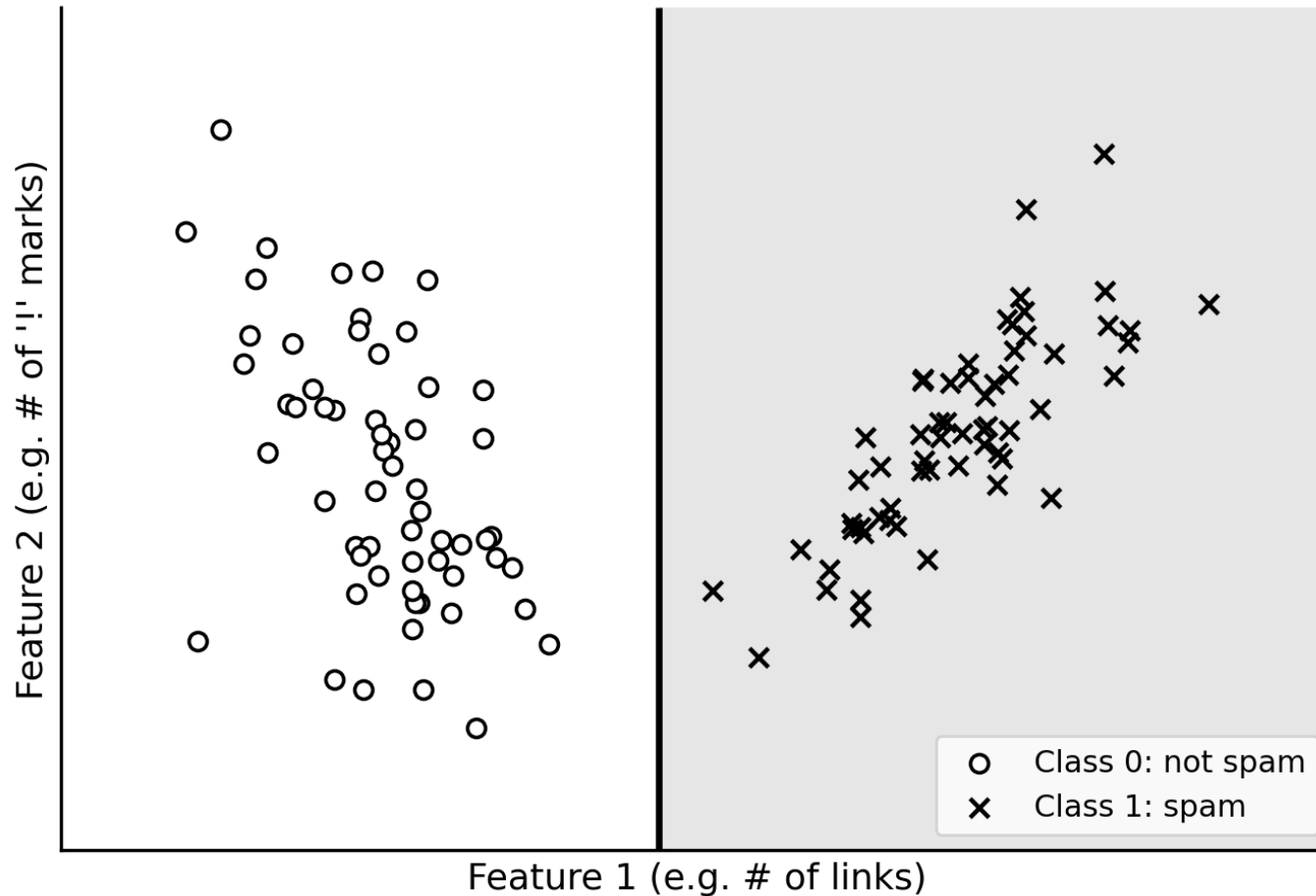
- Spam or not spam
- Which digit (0–9)?
- Will the customer leave? yes/no
- Tumour benign or malignant

Model: logistic regression

Metrics: accuracy · precision · recall · F1 (Week 5)

Quick test: could the answer be 3.7? → regression. Must it be one of a fixed list? → classification. (Note: “logistic regression” is actually a classifier!)

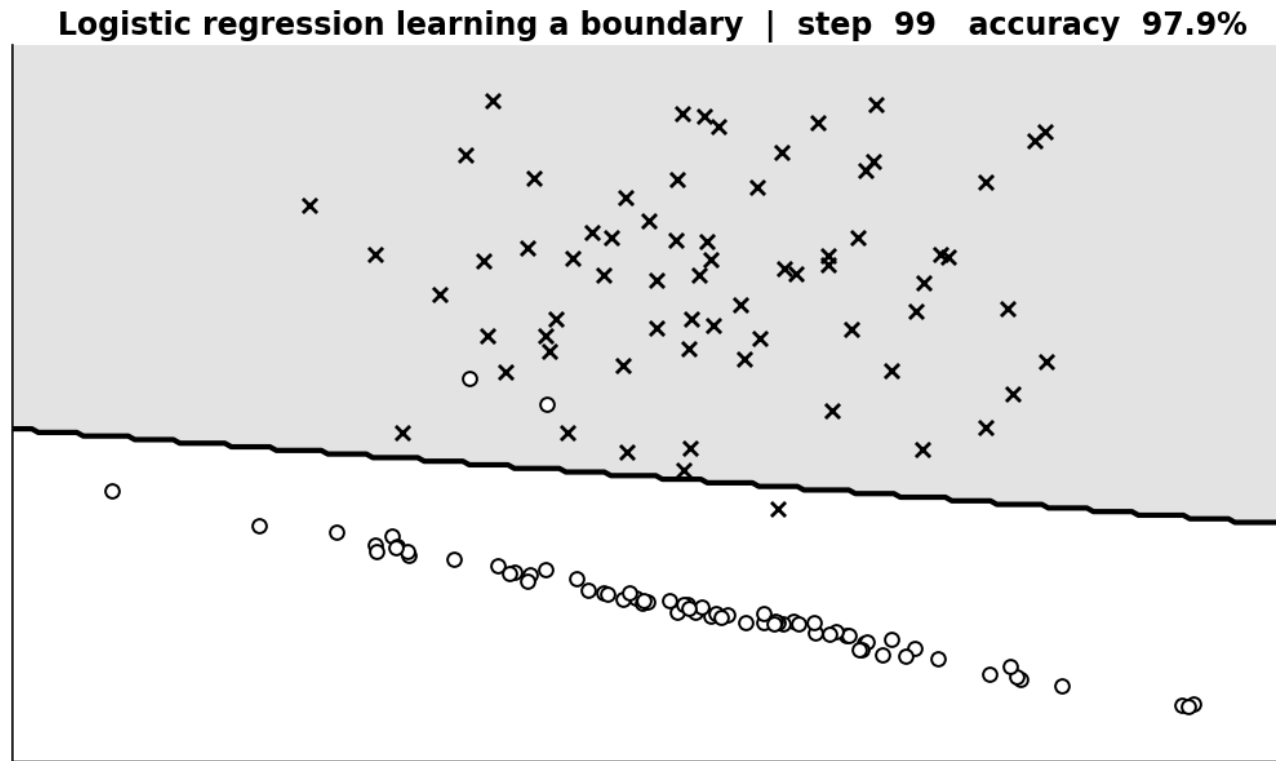
Classification: learning a decision boundary



Synthetic two-feature data. Grey region = predicted spam.

- **Points:** each e-mail is a point described by its features.
- **Boundary:** the line the model learns to separate the classes.
- **Prediction:** a new e-mail gets the class of the side it lands on.
- **Real life:** thousands of features — same idea, higher dimensions.
- **Shapes:** logistic regression → straight; k-NN and trees → flexible.

Video: a classifier learning its boundary

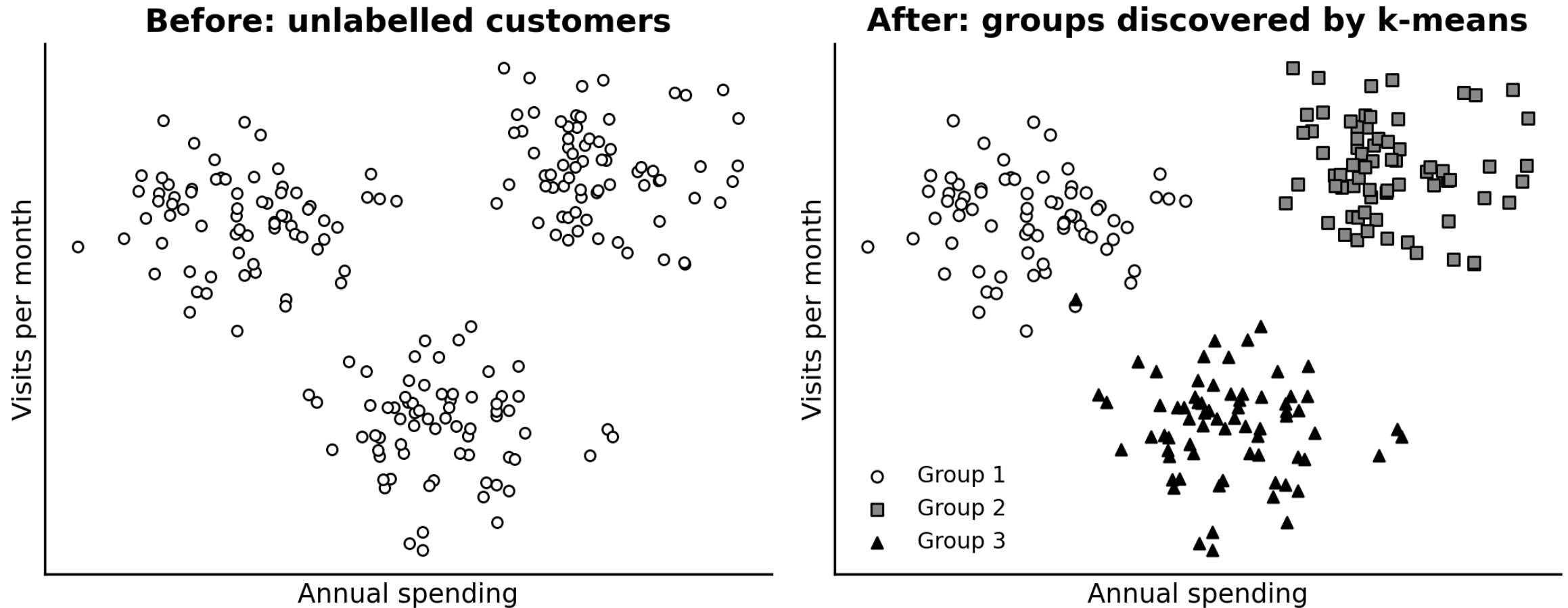


What to watch for

- The boundary starts in the wrong place
- It moves to reduce mistakes
- Accuracy in the title rises
- A few points stay misclassified — that's OK

Click to play (≈7 s). Animation generated for this lecture with Python/matplotlib.

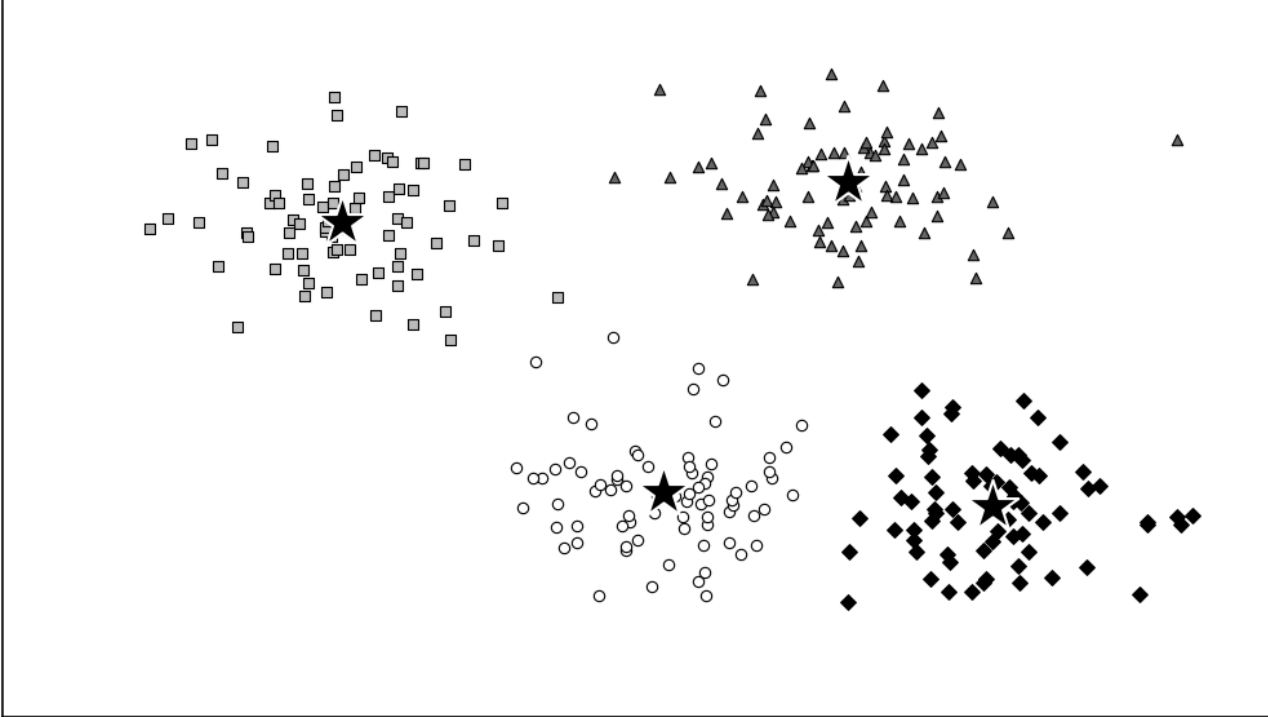
Unsupervised learning: finding hidden structure



Synthetic customer data. Marker shape shows the group found by k-means.

Video: k-means clustering step by step

k-means clustering ($k = 4$) | iteration 8 | move each ★ to its group's centre

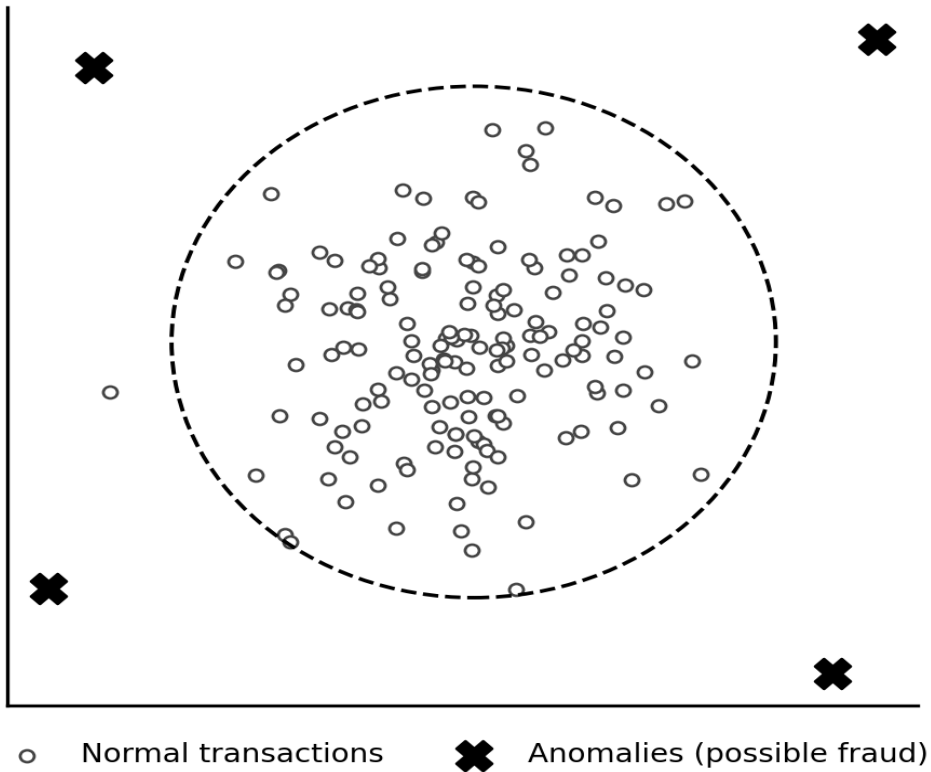


The algorithm

- 1. Place k centres (★)
- 2. Assign each point to its nearest ★
- 3. Move each ★ to the mean of its points
- 4. Repeat 2–3 until nothing changes

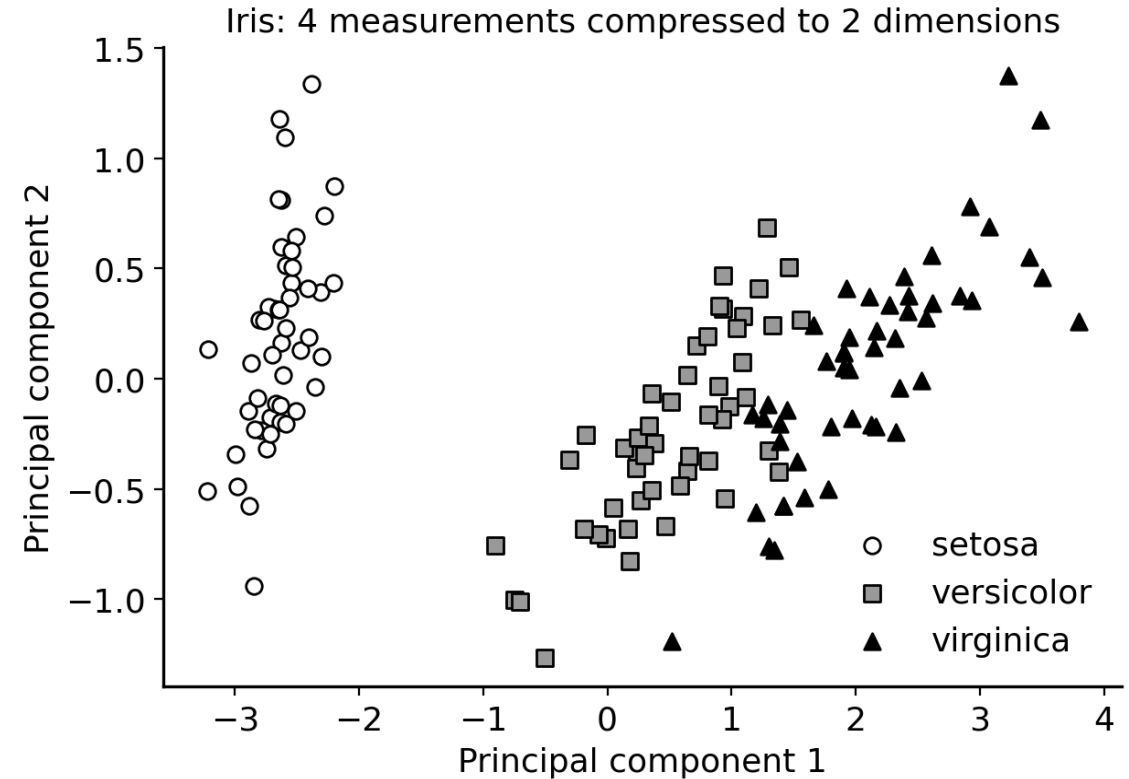
Click to play (≈10 s). Animation generated for this lecture with Python/matplotlib.

Other unsupervised tasks



Anomaly detection

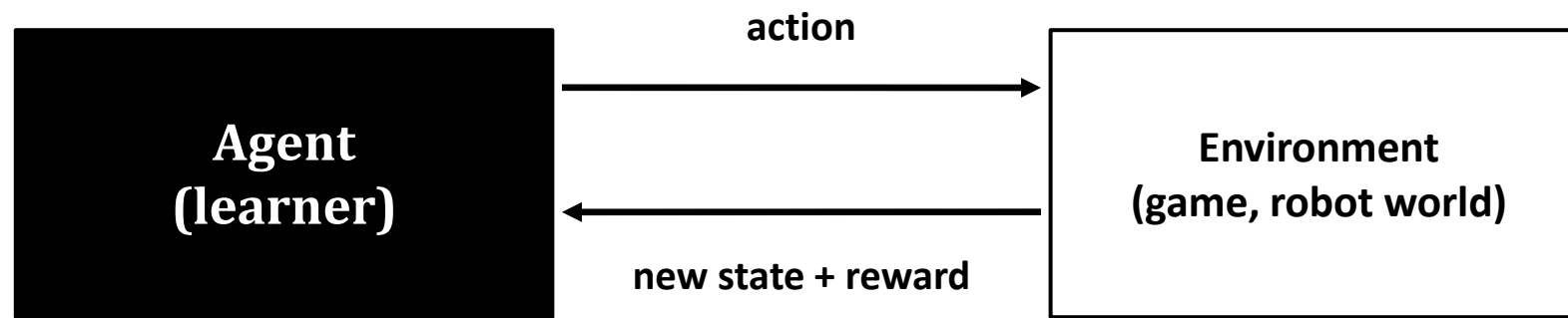
fraud · machine faults · cyber-attacks



Dimensionality reduction

visualise · compress · speed up

Reinforcement learning: learning by trial and error



Goal: learn a policy (a strategy for choosing actions) that maximises the total reward over time.

Analogy: training a dog with treats — no labelled examples, only rewards.



Where it's used

- Games: Go, chess, video games
- Robot control and walking
- Resource management (e.g. cooling)
- Fine-tuning language models with human feedback

Beyond the big three



Semi-supervised

A few labelled examples plus many unlabelled ones.

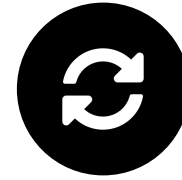
e.g. 1,000 labelled scans + 1,000,000 unlabelled scans



Self-supervised

The data provides its own labels: hide part, predict it.

e.g. predict the missing word — how large language models are pre-trained



Transfer learning

Reuse knowledge from a big task on a new, smaller task.

e.g. general photo model → plant-disease detector with a few hundred images

Quick quiz: which type of ML?

A

Predict next month's electricity use (kWh) of a building from past months.

B

Decide if a skin-lesion photo is benign or malignant, using dermatologist-labelled photos.

C

Group thousands of news articles into topics — no topics are given.

D

A robot arm learns to grasp objects; it gets +1 for each successful grasp.

E

Predict a student's final exam score from attendance and homework marks.

F

Flag unusual log-in attempts to a university system; no attacks are labelled.

Regression · Classification · Unsupervised · Reinforcement? Look for the label first.

Quiz answers

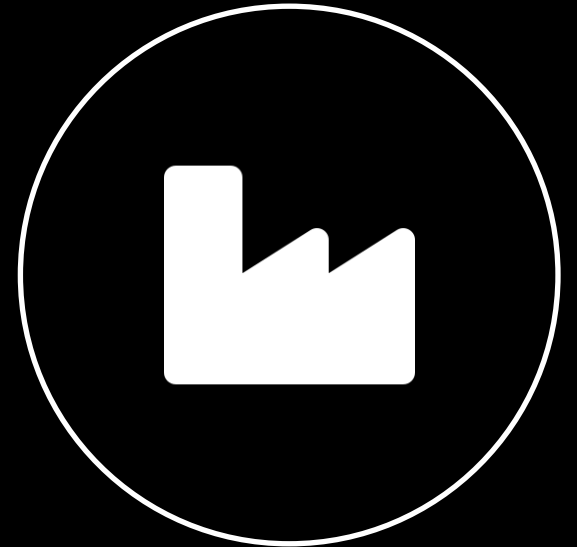
	Type of ML	Why
A	Regression	target is a number (kWh)
B	Classification	benign vs malignant; recall is critical
C	Unsupervised: clustering	no topic labels
D	Reinforcement learning	learns from rewards
E	Regression	a score — or classification if framed as pass/fail
F	Unsupervised: anomaly detection	learn normal, flag unusual

The same data can be framed in different ways — framing the problem is the first and most important decision.

04

ML in the real world

Where machine learning is used today — and where you could use it



Machine learning is everywhere



Healthcare

Diagnosis support from images & ECGs, patient-risk prediction, drug discovery



Finance

Fraud detection in milliseconds, credit scoring, document processing



Retail & online

Recommendations, demand forecasting, pricing



Transport

Arrival-time estimates, route optimisation, driver assistance



Language

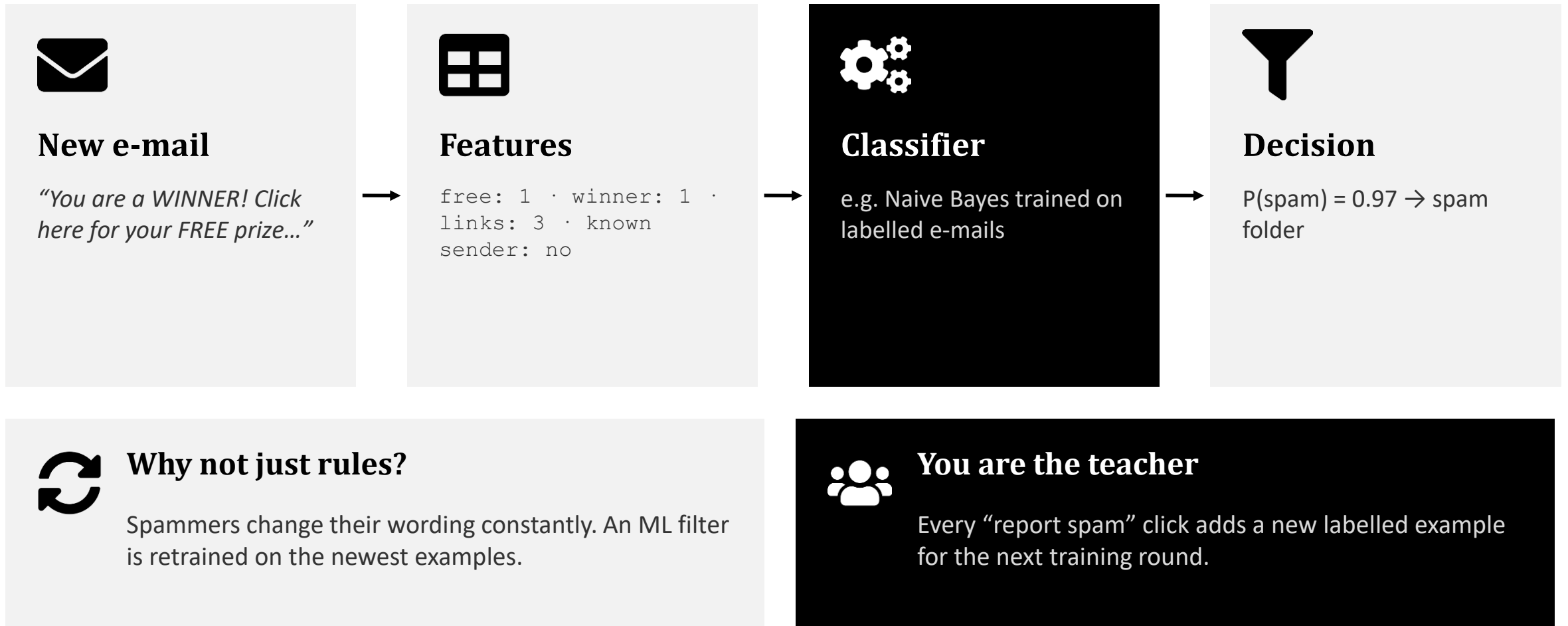
Translation, speech recognition, chatbots, search



Vision

Face unlock, factory quality inspection, crop monitoring

Case study 1: the spam filter



Case study 2: machine learning in healthcare



Where ML helps

- **Medical images:** X-rays, retinal scans, skin photos
- **Biosignals:** ECG and EEG analysis
- **Health records:** predicting readmission risk
- **Workflow:** prioritising urgent cases



Data shift

A model from one hospital or scanner may perform worse at another.



Costly errors

Missing a disease is serious — recall matters more than raw accuracy.



Decision support

ML flags and prioritises; a qualified clinician makes the final decision.

Case study 3: recommender systems

User	Film 1	Film 2	Film 3	Film 4	Film 5
Aziz	5	4	1	?	?
Dilnoza	5	5	1	2	5
Timur	1	2	5	4	?
Malika	?	1	4	5	2

Illustrative ratings (1–5). “?” = not yet rated — the model must predict these.

Aziz $\approx$ Dilnoza on films 1–3 $\rightarrow$ Dilnoza loved film 5 $\rightarrow$ recommend film 5 to Aziz



Collaborative filtering

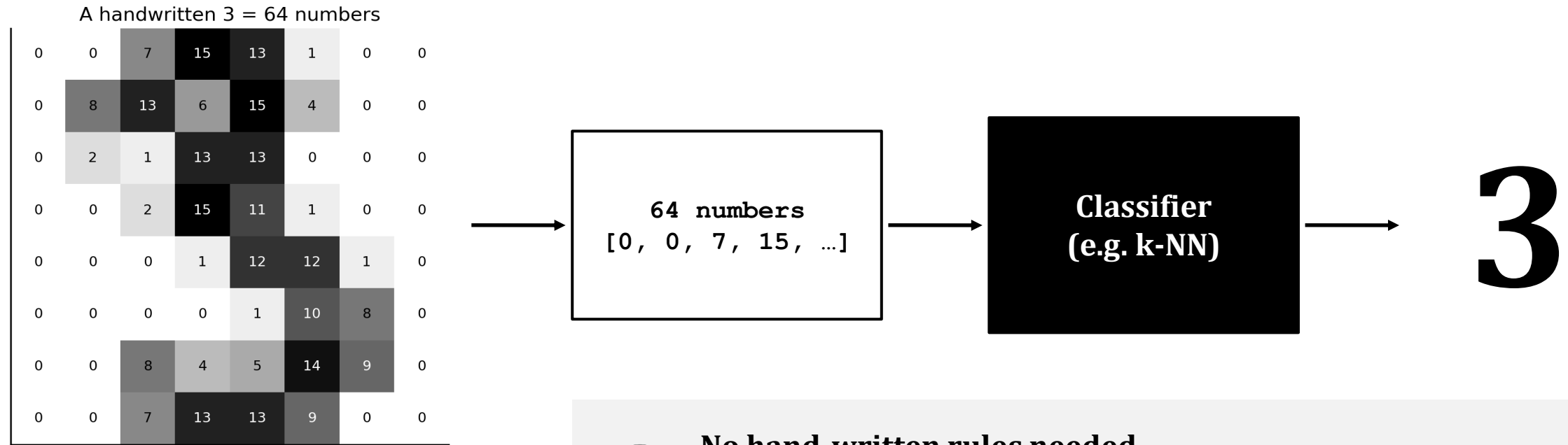
Users who agreed in the past will probably agree in the future.



Everywhere online

Video suggestions, “customers also bought”, music playlists, news feeds.

Case study 4: computer vision — back to our “3”



Input image (8×8)



No hand-written rules needed

Trained on 1,797 labelled images, even simple models typically exceed 95% accuracy on unseen digits. The same idea at scale powers face unlock, postal-code reading and medical imaging.

Project ideas: problems you could solve this semester



Energy

Predict building electricity use from weather and time

Regression



Air quality

Predict pollution levels from weather measurements

Regression



Agriculture

Predict crop yield or classify plant diseases

Regression / classification



Education

Identify students at risk of failing (mind privacy)

Classification



Health

Predict diabetes or heart-disease risk from public data

Classification



Business

Group shop customers by behaviour

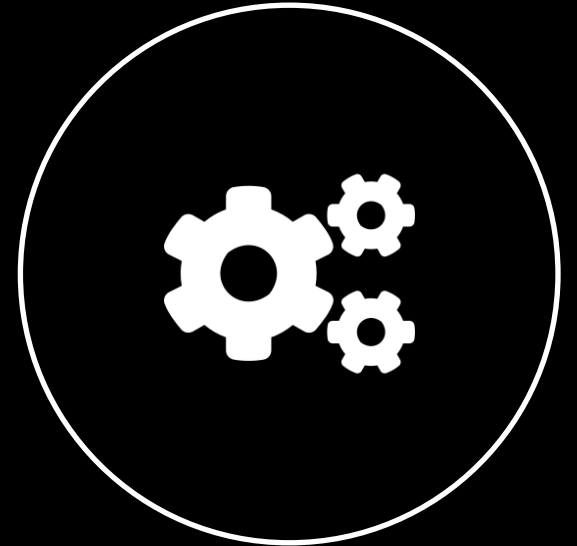
Clustering

Where to find data: Kaggle · UCI Machine Learning Repository · government open-data portals

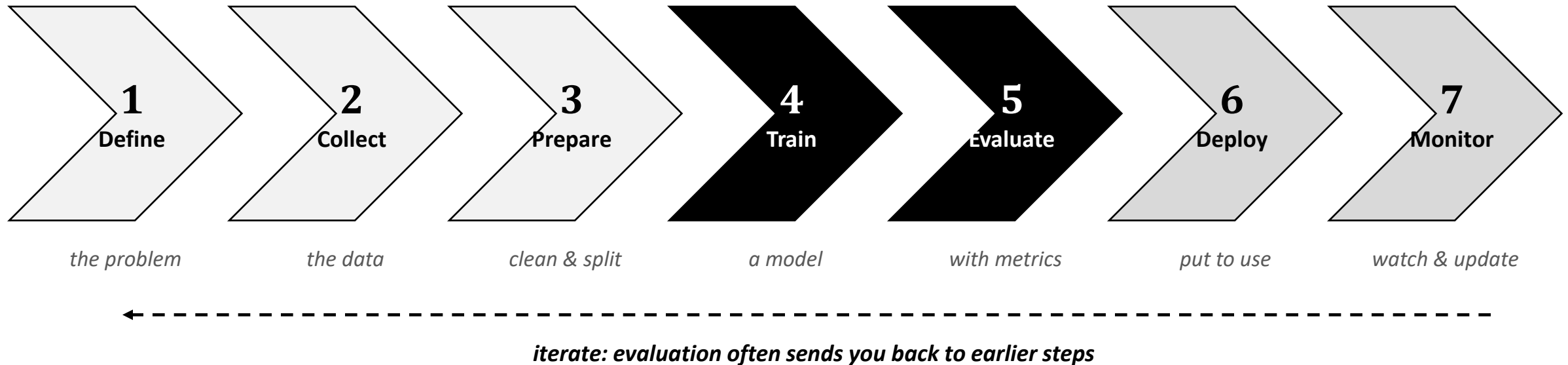
05

The ML workflow

From a question to a working model — plus limits and ethics



The machine learning workflow in seven steps



Steps	Where in this course
1–3 Define · Collect · Prepare	Weeks 1–3 (pandas, missing values, encoding, train/test split)
4–5 Train · Evaluate	Weeks 4–11 (regression, classification, k-NN, trees, Naive Bayes, k-means)
6–7 Deploy · Monitor	Week 12 pipelines; discussed conceptually

Steps 1–2: define the problem and collect data



1 · Define the problem

- What decision will the prediction support?
- What exactly is the target y ?
- Regression, classification or clustering?
- Which metric shows success — and what level is useful?
- What simple baseline must we beat? (e.g. always predict the average)



2 · Collect the data

- Where does it come from? May we use it?
- Is there a clear label column?
- Enough rows? (a few hundred to tens of thousands is ideal here)
- Does it represent real use?
- Does it contain personal data to protect?

Step 3: prepare the data — often the biggest job

Problem	Example	Typical fix	Week
Missing values	Age of house is blank	Drop the row or fill with the median	3
Categorical features	District = “Chilonzor”	One-hot encoding	3
Different scales	Size 40–160 vs rooms 1–4	Standardise / normalise	6
Errors & outliers	Size = -50 m^2 , age = 900 years	Check, correct or remove	2–3
Leakage risk	Scaling on the full dataset	Split first; fit steps on training data only	3, 12

**Garbage in,
garbage out.**

A model can never be better than the data it learns from. Careful preparation is where good projects are won.

Steps 4–5: train and evaluate

Task	Metric	Question it answers
Regression	MAE	On average, how far off are we (in target units)?
Regression	RMSE	Like MAE, but punishes big errors more
Classification	Accuracy	What fraction of predictions are correct?
Classification	Precision	Of the items we flagged, how many were right?
Classification	Recall	Of all real positives, how many did we catch?
Classification	F1-score	A balance of precision and recall

Start simple: first beat a baseline — predict the average (regression) or the most common class (classification).

Actual: not spam	48 True Neg	4 False Pos
Actual: spam	6 False Neg	42 True Pos
	Pred: not spam	Pred: spam

*Example confusion matrix for a spam filter (100 test e-mails).
Accuracy = $(48 + 42) / 100 = 90\%$.*

Steps 6–7: deploy and monitor



Deploy

- Inside a web or mobile app
- Behind an API
- A nightly scoring job
- A dashboard for decision makers



Drift happens

- Customer behaviour changes
- New spam tactics appear
- A hospital buys a new scanner
- Prices rise with inflation



Monitor & retrain

- Track live performance
- Watch incoming data distributions
- Retrain on fresh data
- Keep versions of models and data

In this course, “deployment” = a clean, reproducible notebook/pipeline that anyone can run from top to bottom and get the same results (IWS 3, Week 12; capstone rubric).

When should — and shouldn't — you use ML?



Good fit for ML

- The pattern is too complex for hand-written rules
- The rules change over time (spam, fraud)
- Plenty of relevant, representative data
- Some errors are acceptable
- The task is repeated many times



Poor fit for ML

- A simple rule or formula already works (tax, unit conversion)
- Little or no data
- Every decision must be fully explained or 100% correct
- Errors are dangerous and no human checks the output

Limitations and risks of machine learning



Data quality

Wrong labels or unrepresentative samples produce wrong models.



Bias

Models can learn and repeat unfair patterns in historical data.



Overfitting

Memorising training data instead of learning the pattern.



Explainability

Some models are black boxes — hard to justify decisions.



Correlation ≠ causation

Ice-cream sales and drownings both rise in summer.

Habit for your projects: for every model, write 1–2 sentences on what could go wrong and for whom. (Limitations = 10% of the midterm rubric.)

Responsible and ethical machine learning



Fairness

Does the model work equally well for all groups?



Transparency

Can we explain how it reaches decisions?



Privacy

Minimal, protected, anonymised personal data.



Accountability

A human stays responsible for decisions.

In your projects

- Use public or properly anonymised datasets
- Never include names, phone or ID numbers
- Discuss possible bias in your limitations
- Cite datasets, libraries and adapted code
- Academic-integrity rules apply to code too

06

Tools & what's next

The Python toolkit, your first ML program, the lab and your projects



Your machine learning toolkit



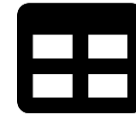
Python 3

The main language of ML



Jupyter / Colab

Notebooks: code + text + charts



NumPy

Fast arrays and maths



pandas

DataFrames for data work



matplotlib

Plots and charts



scikit-learn

Models, splits and metrics



GitHub

Version control & submission



Kaggle / UCI

Datasets to practise on

Your first machine learning program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

X, y = load_iris(return_X_y=True)                # 1 data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)        # 2 split
model = KNeighborsClassifier(n_neighbors=5)       # 3 model
model.fit(X_train, y_train)                      # 4 train
y_pred = model.predict(X_test)                  # 5 predict
print("Accuracy:", accuracy_score(y_test, y_pred)) # 6 evaluate
```

Output

Accuracy: 1.0



Don't celebrate yet

Iris is small, clean and famously easy. On real data, a perfect score usually signals leakage or overfitting.

The scikit-learn pattern: fit, predict, score

1 Create

```
model = KNeighborsClassifier(n_neighbors=5)
```

choose model & hyperparameters

2 Fit

```
model.fit(X_train, y_train)
```

learn from training data

3 Predict

```
y_pred = model.predict(X_test)
```

apply to new data

4 Score

```
model.score(X_test, y_test)
```

quick default metric

Swap models in one line: KNeighborsClassifier → DecisionTreeClassifier → GaussianNB; the rest stays the same. (k-means: fit(X) only.)

Today's lab (LC 1): tools, notebooks, datasets, projects

A



Set up Colab

Create a notebook and save it to Drive

B



Notebook basics

Cells, shortcuts, Restart & run all

C



Library check

NumPy, pandas, matplotlib, scikit-learn

D



Explore datasets

Iris + a CSV from the web; first plot

E

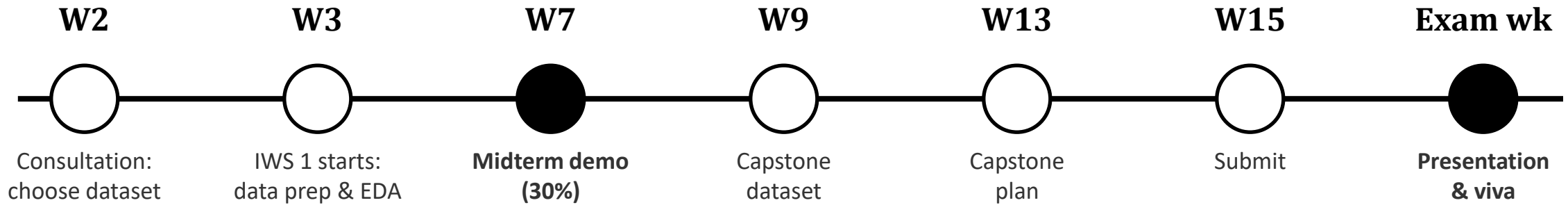


First model & projects

Run the first ML program; judge datasets

Bring: a laptop (if possible) and your Google account login. **Submit:** your completed Lab 1 notebook (.ipynb) by the deadline given in class.

Project expectations: what's coming



Midterm rubric (Week 7)

- Problem understanding 30%
- Model choice 30%
- Evaluation 30%
- Limitations 10%



Capstone rubric

- Problem, models & evaluation 25%
- Workflow & implementation 25%
- Report & notebook clarity 20%
- Presentation & viva 15%
- Teamwork 15%



My advice

- Form your group early
- Pick data you find interesting
- Use all 7 consultations
- Commit to GitHub regularly

Watch and read before next week

WATCH



StatQuest: A Gentle Introduction to Machine Learning

Core ideas, very simple examples



CGP Grey: How Machines Learn

Animated story of how learning systems are built



3Blue1Brown: But what is a neural network?

Optional — a first look at deep learning



AlphaGo – The Movie (DeepMind)

Optional — full documentary



READ

Géron (2022)

Hands-On Machine Learning, 3rd ed. — Ch. 1, The Machine Learning Landscape

James et al. (2023)

An Introduction to Statistical Learning with Applications in Python — Ch. 1–2

Müller & Guido (2016)

Introduction to Machine Learning with Python — Ch. 1

Explore

scikit-learn.org · kaggle.com/datasets

Key takeaways

1

ML learns rules from data and answers — instead of hand-written rules.

2

Learning = adjusting a model's parameters to reduce its error.

3

The real goal is generalisation: test on unseen data, beware overfitting.

4

Three types: supervised (regression, classification), unsupervised, reinforcement.

5

ML is a workflow — and data quality usually matters more than the algorithm.

6

Every model has limitations; build and use ML responsibly.

Machine learning is learning from examples to make good predictions on new data.

Exit ticket (3 minutes)

1

Explain the difference between traditional programming and machine learning in one or two sentences.

2

Name a problem from your own life or interests. Is it regression, classification or clustering? What is the label?

3

Why must we evaluate a model on data it has not seen during training?



Not graded

Write on paper or post in the course MS Teams channel. It helps me plan next week's lecture.

Thank you

Questions?

Next: Lab 1 — course tools, Jupyter notebooks, datasets and project expectations

Week 2: Python for simple data work — pandas basics

Contact: Behnam Kiani · Kiani-bme@hotmail.com · office hours by appointment

